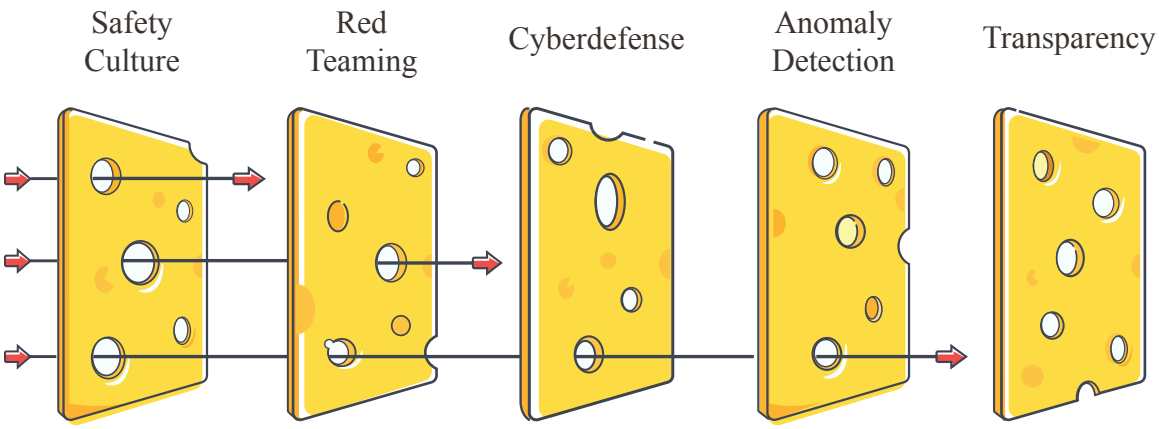

Figure 14: The Swiss cheese model shows how technical factors can improve organizational safety. Multiple layers of defense compensate for each other's individual weaknesses, leading to a low overall level of risk.

Empirical measurement of both safety and capabilities is needed to establish that a safety intervention reduces overall AI risk. Improving a facet of an AI's safety often does *not* reduce overall risk, as general capabilities advances can often improve specific safety metrics. To reduce overall risk, a safety metric needs to be improved relative to general capabilities. Both of these quantities need to be empirically measured and contrasted. Currently, most organizations proceed by gut feeling, appeals to authority, and intuition to determine whether a safety intervention would reduce overall risk. By objectively evaluating the effects of interventions on safety metrics and capabilities metrics together, organizations can better understand whether they are making progress on safety relative to general capabilities.

Fortunately, safety and general capabilities are not identical. More intelligent AIs may be more knowledgeable, clever, rigorous, and fast, but this does not necessarily make them more just, power-averse, or honest—an intelligent AI is not necessarily a beneficial AI. Several research areas mentioned throughout this document improve safety relative to general capabilities. For example, improving methods to detect dangerous or undesirable behavior hidden inside AI systems do not improve their general capabilities, such the ability to code, but they can greatly improve safety. Research that empirically demonstrates an improvement of safety relative to capabilities can reduce overall risk and help avoid inadvertently accelerating AI development, fueling competitive pressures, or hastening the onset of existential risks.

Safetywashing can undermine genuine efforts to improve AI safety. Organizations should be wary of "safetywashing"—the act of overstating or misrepresenting one's commitment to safety by exaggerating the effectiveness of "safety" procedures, technical methods, evaluations, and so forth. This phenomenon takes on various forms and can contribute to a lack of meaningful progress in safety research. For example, an organization may publicize their dedication to safety while having a minimal number of researchers working on projects that truly improve safety.

Misrepresenting capabilities developments as safety improvements is another way in which safetywashing can manifest. For example, methods that improve the reasoning capabilities of AI systems could be advertised as improving their adherence to human values—since humans might prefer the reasoning to be correct—but would mainly serve to enhance general capabilities. By framing these advancements as safety-oriented, organizations may mislead others into believing they are making substantial progress in reducing AI risks when in reality, they are not. It is crucial for organizations to accurately represent their research to promote genuine safety and avoid exacerbating risks through safetywashing practices.

In addition to human factors, safe design principles can greatly affect organizational safety. One example of a safe design principle in organizational safety is the Swiss cheese model (as shown in Figure 14), which is applicable in various domains, including AI. The Swiss cheese model employs a multilayered approach

to enhance the overall safety of AI systems. This "defense in depth" strategy involves layering diverse safety measures with different strengths and weaknesses to create a robust safety system. Some of the layers that can be integrated into this model include safety culture, red teaming, anomaly detection, information security, and transparency. For example, red teaming assesses system vulnerabilities and failure modes, while anomaly detection works to identify unexpected or unusual system behavior and usage patterns. Transparency ensures that the inner workings of AI systems are understandable and accessible, fostering trust and enabling more effective oversight. By leveraging these and other safety measures, the Swiss cheese model aims to create a comprehensive safety system where the strengths of one layer compensate for the weaknesses of another. With this model, safety is not achieved with a monolithic airtight solution, but rather with a variety of safety measures.

In summary, weak organizational safety creates many sources of risk. For AI developers with weak organizational safety, safety is merely a matter of box-ticking. They do not develop a good understanding of risks from AI and may safetywash unrelated research. Their norms might be inherited from academia ("publish or perish") or startups ("move fast and break things"), and their hires often do not care about safety. These norms are hard to change once they have inertia, and need to be addressed with proactive interventions.

Story: Weak Safety Culture

An AI company is considering whether to train a new model. The company's Chief Risk Officer (CRO), hired only to comply with regulation, points out that the previous AI system developed by the company demonstrates some concerning capabilities for hacking. The CRO says that while the company's approach to preventing misuse is promising, it isn't robust enough to be used for much more capable AIs. The CRO warns that based on limited evaluation, the next AI system could make it much easier for malicious actors to hack into critical systems. None of the other company executives are concerned, and say the company's procedures to prevent malicious use work well enough. One mentions that their competitors have done much less, so whatever effort they do on this front is already going above and beyond. Another points out that research on these safeguards is ongoing and will be improved by the time the model is released. Outnumbered, the CRO is persuaded to reluctantly sign off on the plan.

A few months after the company releases the model, news breaks that a hacker has been arrested for using the AI system to try to breach the network of a large bank. The hack was unsuccessful, but the hacker had gotten further than any other hacker had before, despite being relatively inexperienced. The company quickly updates the model to avoid providing the particular kind of assistance that the hacker used, but makes no fundamental improvements.

Several months later, the company is deciding whether to train an even larger system. The CRO says that the company's procedures have clearly been insufficient to prevent malicious actors from eliciting dangerous capabilities from its models, and the company needs more than a band-aid solution. The other executives say that to the contrary, the hacker was unsuccessful and the problem was fixed soon afterwards. One says that some problems just can't be foreseen with enough detail to fix prior to deployment. The CRO agrees, but says that ongoing research would enable more improvements if the next model could only be delayed. The CEO retorts, "That's what you said the last time, and it turned out to be fine. I'm sure it will work out, just like last time."

After the meeting, the CRO decides to resign, but doesn't speak out against the company, as all employees have had to sign a non-disparagement agreement. The public has no idea that concerns have been raised about the company's choices, and the CRO is replaced with a new, more agreeable CRO who quickly signs off on the company's plans.

The company goes through with training, testing, and deploying its most capable model ever, using its existing procedures to prevent malicious use. A month later, revelations emerge that terrorists have managed to use the system to break into government systems and steal nuclear and biological secrets, despite the safeguards the company put in place. The breach is detected, but by then it is too late: the dangerous information has already proliferated.

4.3 Suggestions

We have discussed how accidents are inevitable in complex systems, how they could propagate through those systems and result in disaster, and how organizational factors can go a long way toward reducing the risk of catastrophic accidents. We will now look at some practical steps that organizations can take to improve their overall safety.

Red teaming. Red teaming is a term used across industries to refer to the process of assessing the security, resilience, and effectiveness of systems by soliciting an adversarial "red" team to identify problems [103]. AI labs should commission external red teams to identify hazards in their AI systems to inform deployment decisions. Red teams could demonstrate dangerous behaviors or vulnerabilities in monitoring systems intended to prevent disallowed use. Red teams can also provide indirect evidence that an AI system might be unsafe; for example, demonstrations that smaller AIs are behaving deceptively might indicate that larger AIs are also deceptive but better at evading detection.

Affirmative demonstration of safety. Companies should have to provide affirmative evidence for the safety of their development and deployment plans before they can proceed. Although external red teaming might be useful, it cannot uncover all of the problems that companies themselves might be able to, and is thus inadequate [104]. Since hazards may arise from system training, companies should have to provide a positive argument for the safety of their training and deployment plans before training can begin. This would include grounded predictions regarding the capabilities the new system would be likely to have, plans for how monitoring, deployment, and information security will be handled, and demonstrations that the procedures used to make future company decisions are sound. Just as one does not need evidence that "a gun in loaded to avoid playing Russian roulette, or evidence that a thief is on the lookout to lock your door," [105] the burden of proof should be on the developers of advanced AIs.

Deployment procedures. AI labs should acquire information about the safety of AI systems before making them available for broader use. One way to do this is to commission red teams to find hazards before AI systems are promoted to production. AI labs can execute a "staged release": gradually expanding access to the AI system so that safety failures are fixed before they produce widespread negative consequences [106]. Finally, AI labs can avoid deploying or training more powerful AI systems until currently deployed AI systems have proven to be safe over time.

Publication reviews. AI labs have access to potentially dangerous or dual-use information such as model weights and research intellectual property (IP) that would be dangerous if proliferated. An internal review board could assess research for dual-use applications to determine whether it should be published. To mitigate malicious and irresponsible use, AI developers should avoid open-sourcing the most powerful systems and instead implement structured access, as described in the previous section.

Response plans. AI labs should have plans for how they respond to security incidents (e.g. cyberattacks) and safety incidents (e.g. AIs behaving in an unintended and destructive manner). Response plans are common practice for high reliability organizations (HROs). Response plans often include identifying potential risks, detailing steps to manage incidents, assigning roles and responsibilities, and outlining communication strategies [107].

Internal auditing and risk management. Adapting from common practice in other high-risk industries such as the financial and medical industries, AI labs should employ a chief risk officer (CRO), namely a senior executive who is responsible for risk management. This practice is commonplace in finance and medicine and can help to reduce risk [108]. The chief risk officer would be responsible for assessing and mitigating risks associated with powerful AI systems. Another established practice in other industries is having an internal